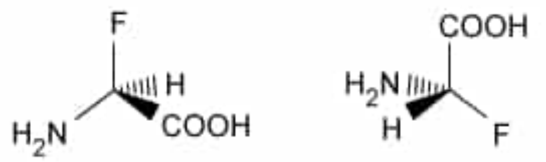


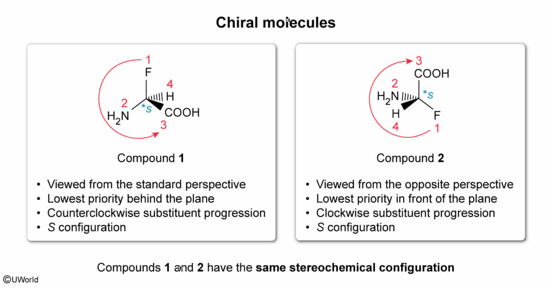
Compound **1** Compound **2**

Consider Compounds **1** and **2** above. Do the compounds have the same stereochemical configuration?

- ✔ A. Yes, because the stereocenters in Compounds **1** and **2** are both *S* [63%]
- B. No, because Compounds **1** and **2** are enantiomers [14%]
- C. Yes, because the stereocenters in Compounds **1** and **2** are both *R* [8%]
- D. No, because Compounds **1** and **2** are diastereomers [13%]

Correct
63% answered correctly

Explanation:



A chiral molecule contains at least one stereocenter and is not superimposable on its mirror image. To determine whether two chiral molecules have the same configuration, the priority of the substituents is assigned and then an *R* or *S* configuration is assigned to the stereocenter.

A stereocenter has an *R* configuration if the **lowest priority** substituent points *behind the plane* (ie, on a **dash**) and priorities 1-2-3 follow a **clockwise** progression; a **counterclockwise** progression indicates an ***S* configuration**. Alternatively, if the **lowest priority** group points *in front of the plane* (ie, on a **wedge**), the molecule is being viewed from the *opposite perspective*, and the stereocenter has an *R* configuration if priorities 1-2-3 follow a counterclockwise progression; a **clockwise** progression indicates an ***S* configuration**.

The priority of the substituents in Compounds **1** and **2** (ranked from highest to lowest) is F, NH_2, COOH, and H. In Compound **1**, the lowest priority group points behind the plane and priorities 1-2-3 follow a counterclockwise progression, indicating an *S* configuration. In Compound **2**, the lowest priority group points in front of the plane and priorities 1-2-3 follow a clockwise progression, indicating an *S* configuration. Therefore, Compounds **1** and **2** have the same stereochemical configuration.

(Choice B) Compounds **1** and **2** would be enantiomers if the configurations of the stereocenters were *opposite*.

(Choice C) Compound **1** would have an *R* configuration if priorities 1-2-3 followed a *clockwise* progression with the lowest-priority substituent pointing behind the plane. Compound **2** would have an *R* configuration if priorities 1-2-3 followed a *counterclockwise* progression with the lowest-priority substituent pointing in front of the plane.

(Choice D) Compounds **1** and **2** cannot be diastereomers because these molecules contain only one stereocenter and diastereomers have at least two stereocenters.

Educational objective:
When the lowest-priority substituent in a chiral molecule points behind the plane, a clockwise progression of substituent priorities indicates a stereocenter with an *R* configuration and a counterclockwise progression indicates an *S* configuration. If the lowest-priority substituent points in front of the plane, the molecule is viewed from the opposite perspective and these indications are observed backward.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403209
System:
Introduction to Organic Chemistry